Sujit Debnath

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OBJECTIVE

Enthusiastic, highly-motivated Computer Science graduate with proven teaching, analysis and volunteering competencies who likes to take initiatives and seeks new challenges. I have great interest in the field of Machine Learning and Data Science particularly devising a better problem-solving method for complicated tasks while learning new technologies and tools.

EDUCATION

North South University — **Bashundhara R/A, Dhaka**
Bachelor of Science in Computer Science and Engineering; CGPA: 3.66/4.00 — *Jan 2015 – Dec 2018*

Dhaka College — **Dhanmondi, Dhaka**
Higher Secondary Certificate; Science Group; GPA: 5.00/5.00 — *2011 – 2013*

Rampura Ekramunnesa Boys' High School — **Rampura, Dhaka**
Secondary School Certificate; Science Group; GPA: 5.00/5.00 — *2009 – 2011*

EXPERIENCE

Robi Axiata Limited — **Gulshan, Dhaka**
***Project Coordinator**, Department of Analytics Center of Excellence* — *February 2020 – Present*

- Core responsibilities are to check and analyze the Key Performance Indicators (KPIs) from the vast amount of data of newly upgraded sites and share feedback with the corresponding departments or vendors (e.g., Huawei, Ericsson).
- Currently, working on a project whose goal is to build a custom conversational chatbot that can interact with group members of *Microsoft Kaizala*, follow instructions, and send feedback accordingly.

North South University — **Bashundhara R/A, Dhaka**
***Lab Officer**, Department of Electrical & Computer Engineering* — *Spring 2019 – Fall 2019*

- **Conduct lab classes for the following courses:**
  - CSE 115L (Programming Language I Lab): Summer 2019, Fall 2019
  - CSE 215L (Programming Language II Lab): Summer 2019, Spring 2019, Fall 2019
  - CSE 225L (Data Structures and Algorithms Lab): Fall 2019
  - CSE 311L (Database Systems Lab): Spring 2019
- **Responsibilities:**
  - Designing course curriculum and conducting lab classes
  - Arrange and evaluate assignments, exams, and projects
  - Invigilate during exams and assist for grading

North South University — **Bashundhara R/A, Dhaka**
***Under-Graduate Assistant**, Department of Electrical & Computer Engineering* — *Spring 2019*

- **Assist Faculty members for the following course:** CSE 115 (Programming Language I), Instructors- *Dr. A. K. M. Ashikur Rahman* & *Dr. Muhammad Asif Hossain Khan*
- **Responsibilities:**
  - Conduct tutorial sessions for students needing extra help outside of class hours
  - Grading of home-works, and assignments
  - Assist faculty members in their course-related work

PUBLICATIONS

- Shakil Ahmed Sumon, Joydip Chowdhury, **Sujit Debnath**, Nabeel Mohammed, Sifat Momen, **"Bangla Short Speech Commands Recognition Using Convolutional Neural Networks"**, In *1st International Conference on Bangla Speech and Language Processing* (*ICBSLP*), pp. 1-6, 2018.
- Nazmul Alam Diptu, Md. Asif Khan, **Sujit Debnath**, Abdullah Al Imam, Al Mahadi Hasan Rakib, Kazi Asfaq Ahmed Ador, Rashedur M. Rahman, **"Early Detection of Glaucoma Using Fuzzy Logic in Bangladesh Context"**, In *9th International Conference on Intelligent Systems* (*IS'18*), pp. 87-93, 2018.

PROJECTS

- **Kothamala - A Bangla Speech Recognition Engine:** This research-based project was conducted in two divisions; one was Bengali Short Speech Commands Recognition, and the other one was Bengali Transcription. In the first division, a robust dataset had been collected with 50 classes, and three different Convolutional Neural Network (CNN) architectures had been proposed. In contrast, combined with a language model, another three different model architectures with Connectionist Temporal Classification (CTC) loss function had been experimented for building speech to text engine, where an open-source dataset was taken from *OpenSLR*, released by Google.
- **Early Risk Prediction of Stroke:** In this research-based project, several Machine Learning techniques such as Logistic Regression, Support Vector Classifier, etc along with Synthetic Minority Over-sampling Technique (SMOTE) had been used to predict the risk of stroke on 16 different features excluding the class.

SKILLS

- **Programming Languages:** Python (Frequently used), JAVA, C, C++, SQL
- **Machine Learning, Deep Learning and Visualization Libraries:**
  - Python based libraries: Numpy, Pandas, SciPy
  - Machine Learning library: Scikit-learn
  - Deep Learning library: Keras
  - Visualization: Matplotlib, Plotly, Seaborn
- **Database:** Relational database - MySQL, and NoSQL database - MongoDB
- **Version Control System:** Git
- **Simulation Tools and Others:** Matlab, Logisim
- **Packages:** MS Word, MS PowerPoint, MS Excel and LaTeX.
- **Languages:** Bengali (Native), and English (Proficient).

EXTRACURRICULAR ACTIVITIES

NSU Problem Solvers (NSUPS) **Bashundhara R/A, Dhaka**
Member *2015 – 2016*

- NSUPS is a problem-solving community based on North South University and this community's primary purpose is to help beginner students get involved in competitive programming. However, I got involved with this community at the beginning of my undergraduate life and had the opportunity to participate in several programming competitions organized by NSUPS.

JAAGO Foundation **Banani, Dhaka**
Volunteer *2014 – 2015*

- JAAGO Foundation works for the betterment of the nation through fulfilling the educational needs of under-privileged children. Besides, **Volunteer for Bangladesh (VBD)** is a non-political volunteer and youth wing of JAAGO, which provides the youth of Bangladesh a voice and platform to contribute to nation-building.
- As a volunteer at VBD, Dhaka division, I attended several events organized by JAAGO and VBD. Some notable events were Universal Children's Day, World Environment Day, and The Great Kindness Challenge.

SCHOLARSHIP & ACHIEVEMENTS

- **Financial Aid:** Achieved 25% tuition waiver from North South University *Summer 2015 – Fall 2018*
- **Honors in BS:** *Magna Cum Laude*, Graduating class of 2019, 22nd Convocation of NSU *March 2019*

REFERENCES

- **Dr. Mohammad Rashedur Rahman**
  Professor, Department of Electrical and Computer Engineering, North South University.
  rashedur.rahman@northsouth.edu ece.northsouth.edu/people/rashedur-rahman
- **Dr. Rajesh Palit**
  Associate Professor, Department of Electrical and Computer Engineering, North South University.
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